1 An object undergoes SHM of amplitude A. Through what total distance does the object move during one complete cycle of its motion.(A)A(B)A/2(C)2A(D)4A(E)none of the above

2 An object oscillates with SHM along the Y axis. Its displacement from the origin varies with time according to the equation $Y=4\cos(\pi t+\pi/4)$.Determine the amplitude (A)4m(B)8m(C)2m(D)12m(E)16m

3 In question 2 above, calculate the velocity and acceleration of the object at t=1s (A)2m/s,27.9m/s^2(B)8.89m/s,27.9m/s^2(C)2m/s,8.89m/s^2(D)27.9m/s,8.89m/s^2(E)26.6 m/s,39.5m/s^2

4 In the question 2 above, what is the phase of motion at t=2s (A)20π/4rad(B)15π/4rad(C)9π/4rad(D)5π/4rad(E)0rad

5 A block of mass 200g is hanged by a spring with K=5N/m is free to oscillate on a horizontal frictional surface and displaced 5cm from equilibrium. Find the period of its motion (A) 5.26s (B) 4.26s(C) 3.26s (D) 2.26s (E) 1.26s

6 In the question 5 above, what is the maximum acceleration of the block (A)5.25m/s^2(B)1.26m/s^2(C)0.25m/s^2(D)1.25m/s^2(E)0.05m/s^2

7 A 0.5kg cube connected to a light spring for which the force constant is 20N/m oscillates on a horizontal frictionless track. Calculate the total energy of the system (A)9.0×10^{-3}J(B)2.0×10^{-3}J(C)1.9×10^{-3}J(D)1.9×10^{-4}J(E)9.0×10^{-4}J

8 A traveling wave that cause the particles of the disturbed medium to move perpendicular to the wave motion is called (A) longitudinal wave(B)traveling wave (C)transverse wave(D)wave motion (E)amplitude

9 Which of the following relationships between the acceleration and the displacement X of a particle involve SHM(A)0.5X(B)400X^2(C)-20X(D)-3X^2(E)none of the above.

10 The piston in the cylinder head of a locomotive has a stroke(twice amplitude)of 0.76m.If the piston moves with simple harmonic motion with an angular frequency of 180rev/min,what is its maximum speed(A)5.7m/s(B)7.2m/s(C)8.7m/s(D)5.6m/s(E)5.8m/s

11 Two particles oscillate in SHM along a common straight line segment of length A. Each particle has a period of 1.5s,but they differ in phase by π/6rad.How far apart are they in terms of A,0.5s after the lagging article leaves one end of the path (A)0.183A(B)0.283A(C)0.383A(D)0.483A(E)0.583A

12 In the question 11 above, what is the direction of their motion.(A)towards each other(B)away from each other(C)same direction (D)either towards or away from each other(E)not determinable

13 Find the mechanical energy of a block spring system having a spring constant of 1.3N/cm and amplitude 2.4cm (A)33mJ(B)34mJ(C)35mJ(D)36mJ (E)37mJ

14 Here are the equations of three waves(i)Y=2sin(4x-2t),(ii)Y=sin(3x-4t), Y=2sin(3x-3t) arrange them according to their amplitude, the greatest first. (A)i,ii,iii(B)ii,i,iii(C)ii,iii,i(D)iii,i,ii(E)i,iii,ii

15 The identical sinusoidal waves moving in the same direction along a stretched string interfere with each other. Amplitude is 9.8mm and $\Phi=100^0$.What is the amplitude of the resultant wave.(A)100mm(B)50mm(C)25mm(D)15mm(E)13mm

16 The interference of two identical sinusoidal waves in opposite directions produces (A)sinusoidal waves(B)resonance(C)standing waves(D)progressive waves(E)none of the above

17 A wave has angular frequency of 110rad/s and a wave wavelength of 1.80m.Calculate the angular wave number (A)3.49/m (B)3.59/m (C)4.59/m(D)5.59/m(E)6.59/m

18 A stone is dropped into a well. The sound of the splash is heard 3s later. What is the depth of the well(A)40.7m(B)50.8m(C)90.9m(D)100.2m(E)150.8m

19 A source emits sound wave isotropically.The intensity of the wave 2.5m from the source is 1.9×10^{-4}W/m^2(A)45mW(B)40mW(C)35mW(D)15mW(E)10mW

20 A certain sound source is increased in sound level by 30dB.By what multiple is its intensity increased(A)5000(B)4000(C)3000(D)2000(E)1000

21 Find the speed of waves on a violin string of mass 800mg an length 22cm if f_0=920Hz A)905m/s(B)900m/s(C)405m/s(D)400m/s(E)300m/s

22 A moth at about eye level is 10cm in front of a plane mirror, if you are 30cm from the mirror.What is the distance between your eyes and the apparent position of the moth's image in the mirror.(A)40cm(B)80cm(C)20cm(D)70cm(E)50cm

23 A concave mirror has a radius of curvature of 35cm.It is positioned so that the image of a man's face is 2.5times the size of the face. How far is the mirror from the face? (A)22cm(B)10.5cm(C)20cm(D)10cm(E)18cm

24 An object is 20cm to the left of a thin diverging lens having a 30cm focal length. What is the image distance (A)-24cm(B)-12cm(C)24cm(D)12cm(E)none of the above

25 Monochromatic green light of wavelength 550nm illuminates two parallel narrows slits 7.7μm apart. Calculate the angular deviation of third order (A)0.216rad(B)0.555rad(C)0.781rad(D)0.945rad(E)1.222rad.

26. Calculate the minimum thickness of a soap bubble film (n=1.33) that results in constructive interference in the reflected lights if the film is illuminated with light whose wavelength in free space is $\lambda = 600$ nm .
(A) 133nm (B) 113nm (C) 115nm (D) 789nm (E) 564nm

27. A laser beam ($\lambda = 632.8$nm) is incident on two slits 0.2mm apart. How far apart are the bright interference fringes on a screen 5m away from the slits.
(A) 1.24cm (B) 5.57cm (C) 1.58cm (D) 2.62cm (E) 1. 54cm

28. A viewing screen is separated from a double slit source by 1.2m. The distance between the two slits is 0.03mm. The second order fringe (m=2) is 4.5 cm from the centre line. Determine the wavelength of the light.
(A) 560nm (B) 220nm (C) 140nm (D) 450nm (E) None of the above

29. In question 28 above, calculate the distance between adjacent bright fringes
(A) 2.2cm (B) 1.4cm (C) 4.5cm (D) 5.6cm (E) 1. 3cm

30. The source of light called monochromatic has
(A) Single wavelength (B) Double wavelength (C) Tripple wavelength
(D) All of the above (E) none of the above

31. Assume that a certain spherical mirror has a total length of 10cm. Locate and describe the image for object distance 25cm
(A) 16.7cm, larger than object (B) 16.7cm, smaller than object (C) -10cm, larger than object (D) 10cm, larger than object (E) None of the above

32. A woman who is 1.5m tall is located 3m from antishoplifting mirror. The focal length of the mirror is -0.25m. Find the position of her image.
(A) -10.0cm (B) -7.0cm (C) -5.0cm (D) -0.23cm (E) -0.43cm

33. Find the magnification in question 32 above
(A) 0.01 (B) 0.03 (C) 0.05 (D) 0.07 (E) 0.08

34. In question 32 above, find the height of the image
(A) 0.12m (B) 0.13m (C) 0.14mm (D) 0.15m (E) 0.16m

35. A small fish is swimming at a depth d below the surface of a pond. What is the apparent depth of the fish as viewed from directly overhead n = 1.33
(A) - 0.9d (B) - 0.8d (C) 0.9d (D) 0.8d (E) 0.7d

36. Which of the following equations is a lens maker's equation
(A) $1/p_1 + n/q_1 = n-1/R_1$ (B) $n/p_2 + 1/q_2 = 1 - n/R_2$
(C) $- n/q_1 + 1/q_2 = 1-n/R_2$ (D) $1/f = 1/v + 1/u$
(E) $1/f =(n-1)\{1/R_1 . 1/R_1 \}$

37. What is the focal length of a pane of window glass
A) infinite (B) finite (C) real is positive (D) lateral magnification
(E) None of the above

38. A diverging lens has a focal length of – 20cm. An object 2cm tall is placed 30cm in front of the lens. Locate the image.
(A) 15cm (B) -15cm (C) 12cm (D) -12cm (E) -10cm

39. Determine both the magnification and the height of the image in question 38 above
(A) 2, -10cm (B) 0.4, 0.8cm (C) 0.4, 2cm (D) 0.4,12cm (E) None of the above

40. A converging glass lens (n = 1.52) has a focal length of 40cm in air. Find its focal length when it is immersed in water (n = 1.33)
(A) 150cm (B) 149cm (C) 148cm (D) 147cm (E) 146cm

41. A particular nearsighted person is unable to see objects clearly when they are beyond 2.5m away. What should the focal length be in a lens prescribed to correct this problem.
(A) 00cm (B) -2.5m (C) 2.5m (D) -5.0m (E) 5.0m

42. What is the power of this lens in question 41 above
(A) -0.40 diopter (B) 0.40 diopter (C) - 1 diopter (D) 1 diopter
(E) 0 diopter

43. The ratio of the angle substended by an object with a lens in use to the angle substended by the object placed at the near point with no lens in use is called
(A) diopter (B) angular magnification (C) Lateral magnification (D) Power
(E) None of the above

44. What is the maximum magnification that is possible with a lens having a focal length of 10cm
(A) 2.5 (B) 3.5 (C) 4.5 (D) 5.5 (E) 6.5

45. Greater magnification can be achieved by combining two lenses in a device called
(A) Compound microscope (B) Mirror (C) Periscope (D) Telescope
(E) Kaleidoscope

46. Find the critical angle for a water – air boundary (n = 1.33)
(A) 180^0 (B) 360^0 (C) 48.8^0 (D) 58.8^0 (E) 68.8^0

47. A laser in a compact disc player generates light that has a wavelength of 780nm in air. Find the speed of this light once it enters the plastic of a compact disc (n = 1.55)
(A) 1.94×10^8 mls (B) 1.55×10^8 mls (C) 3.00×10^8 mls
(D) 1.47×10^8 mls (E) 1.92×10^8 mls

48. Find the frequency of the light in air and in the plastic in question 47 above.
(A) 3.85×10^{14} H_z, 1.94×10^{14} H_z (B) 1.94×10^{14} H_z, 3.85×10^{14} H_z
(C) 5.88×10^{14} H_z 1.94×10^{14} H_z (D) 8.85×10^{14} H_z 1.94×10^{14} H_z
(E) 3.85×10^{14} H_z 1.94×10^{14} H_z in both cases

49. A light ray of wavelength 589nm travelling through air is incident on a smooth, flat slab of crown glass at an angle of 30^0 to the normal. Find the angle of refraction.
(A) 30^0 (B) 25^0 (C) 20^0 (D) 19^0 (E) 15^0

50. In question 49 above, use the principle of reversibility of light, determine the angle of refraction.
(A) 60^0 (B) 50^0 (C) 40^0 (D) 38^0 (E) 30^0

(1). A (2). D (3). C (4). B (5). C (6). A (7). E (8). C (9). B (10). C

(11). B (12). A (13). C (14). B (15). A (16). B (17). A (18). C (19). D (20). A

(21). C (22). B (23). A (24). B (25). C (26). C (27). A (28). A (29). C (30). A

(31). A (32). B (33). C (34). D (35). C (36). E (37). B (38). B (39). A (40). C

(41). C (42). A (43). B (44). A (45). A (46). C (47). D (48). A (49). B (50). C